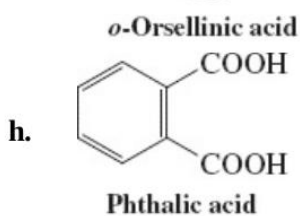
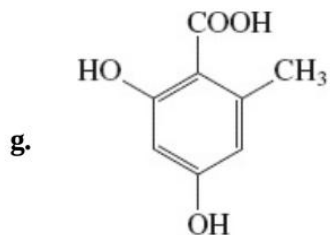
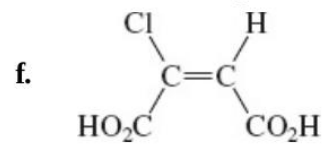
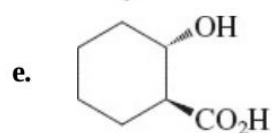
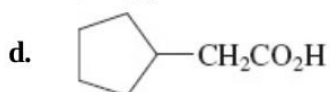
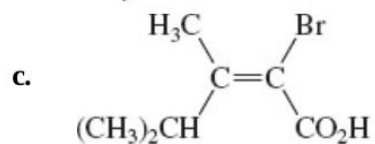
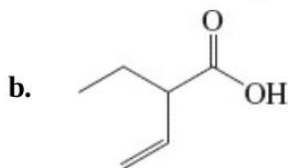
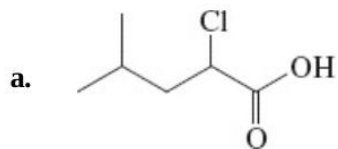
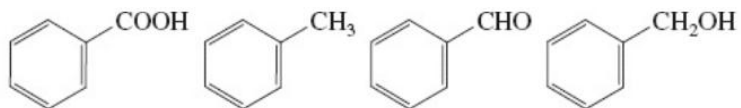


27. Name (IUPAC or *Chemical Abstracts* system) or draw the structure of each of the following compounds.



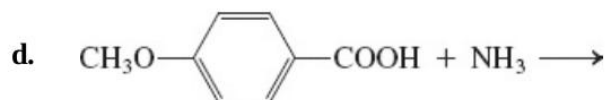
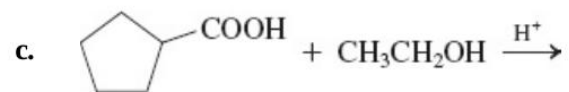
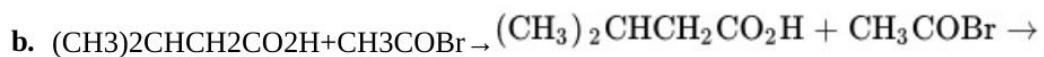
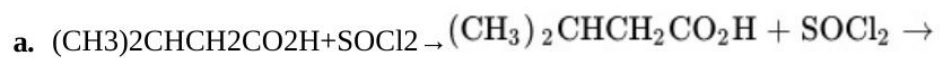
29. Rank the group of molecules shown here in decreasing order of boiling point and solubility in water. Explain your answers.



- 30.** Rank each of the following groups of organic compounds in order of decreasing acidity.

32. **(a)** An unknown A has the formula $C_7H_{12}O_2$ and infrared spectrum A (below). To which class does this compound belong? **(b)** Use the other spectra (NMR-B, p. 961, and F, p. 962; IR-D, E, and F, pp. 961-962) and spectroscopic and chemical information in the reaction sequence to determine the structures of A and the other unknown substances B through F. References are made to relevant sections of earlier chapters, but do not look them up before you have tried to solve the problem without the extra help. **(c)** Another unknown compound, G,

33. Give the products of each of the following reactions.

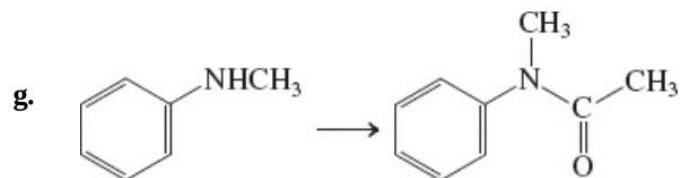
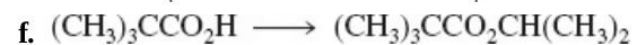
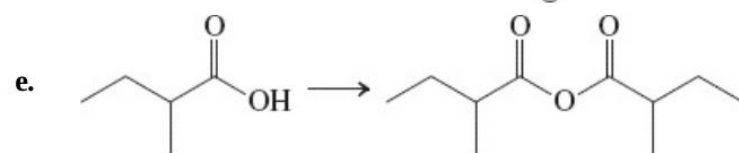
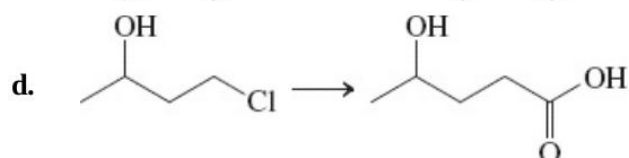
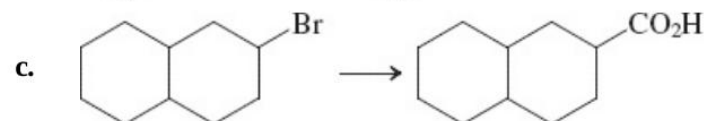
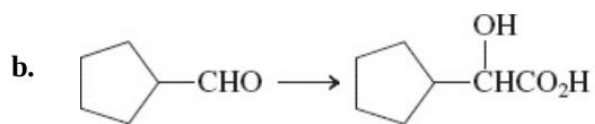
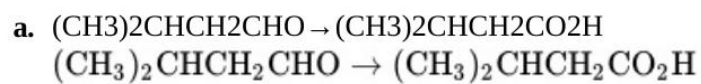


e. Product of (d), heated strongly

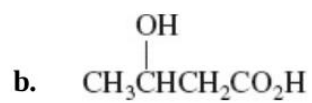
f. Phthalic acid [[Problem 27](#), part (h)], heated strongly

34. When 1,4- and 1,5-dicarboxylic acids, such as butanedioic (succinic) acid ([Section 19-8](#)), are treated with SOCl_2 or PBr_3 in attempted preparations of the diacyl halides, the corresponding cyclic anhydrides are obtained. Explain mechanistically.

37. Fill in suitable reagents to carry out the following transformations.

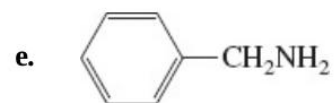
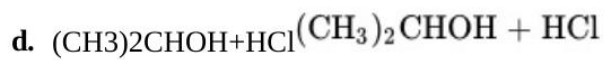


38. Propose syntheses of each of the following carboxylic acids that employ at least one reaction that forms a carbon-carbon bond.

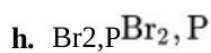
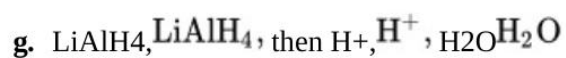


39. (a) Write a mechanism for the esterification of propanoic acid with ^{18}O -labeled ethanol. Show clearly the fate of the ^{18}O label. (b) Acid-catalyzed hydrolysis of an unlabeled ester with ^{18}O -labeled water (H_2^{18}O) leads to incorporation of some ^{18}O into *both* oxygens of the carboxylic acid product. Explain by a mechanism. (**Hint:** You must use the fact that all steps in the mechanism are reversible.)

40. Give the products of reaction of propanoic acid with each of the following reagents.



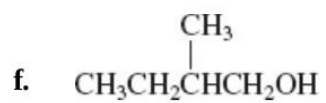
f. Product of (e), heated strongly



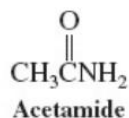
- 41.** Give the product of reaction of cyclopentanecarboxylic acid with each of the reagents in [Problem 40](#).

44. Suggest a preparation of hexanoic acid from pentanoic acid.

45. Give reagents and reaction conditions that would allow efficient conversion of 2-methylbutanoic acid into **(a)** the corresponding acyl chloride; **(b)** the corresponding methyl ester; **(c)** the corresponding ester with 2-butanol; **(d)** the anhydride; **(e)** the *NN*-methanide;



49. (a) (4-Nitrophenyl)ethanenitrile ([Section 19-6, p. 925](#)) is useful as an acid-base indicator in the high pH range. It is yellow below its pK_a of 13.4 and red above. Using the principles that govern acidity, identify the most acidic hydrogen(s) in the molecule, draw the structure of its conjugate base, and provide an explanation for its acidity. (b) Draw resonance forms to explain the stability of protonated propenoic acid ([Section 19-3, p. 919](#)), a common fragment observed in the mass spectra of longer chain carboxylic acids. (c) How would you expect the acidity of acetamide to compare with that of acetic acid? With that of acetone? Which protons in acetamide are the most acidic? Where would you expect acetamide to be protonated by very strong acid?



52. Suggest structures for the products of each reaction in the following synthetic sequence.